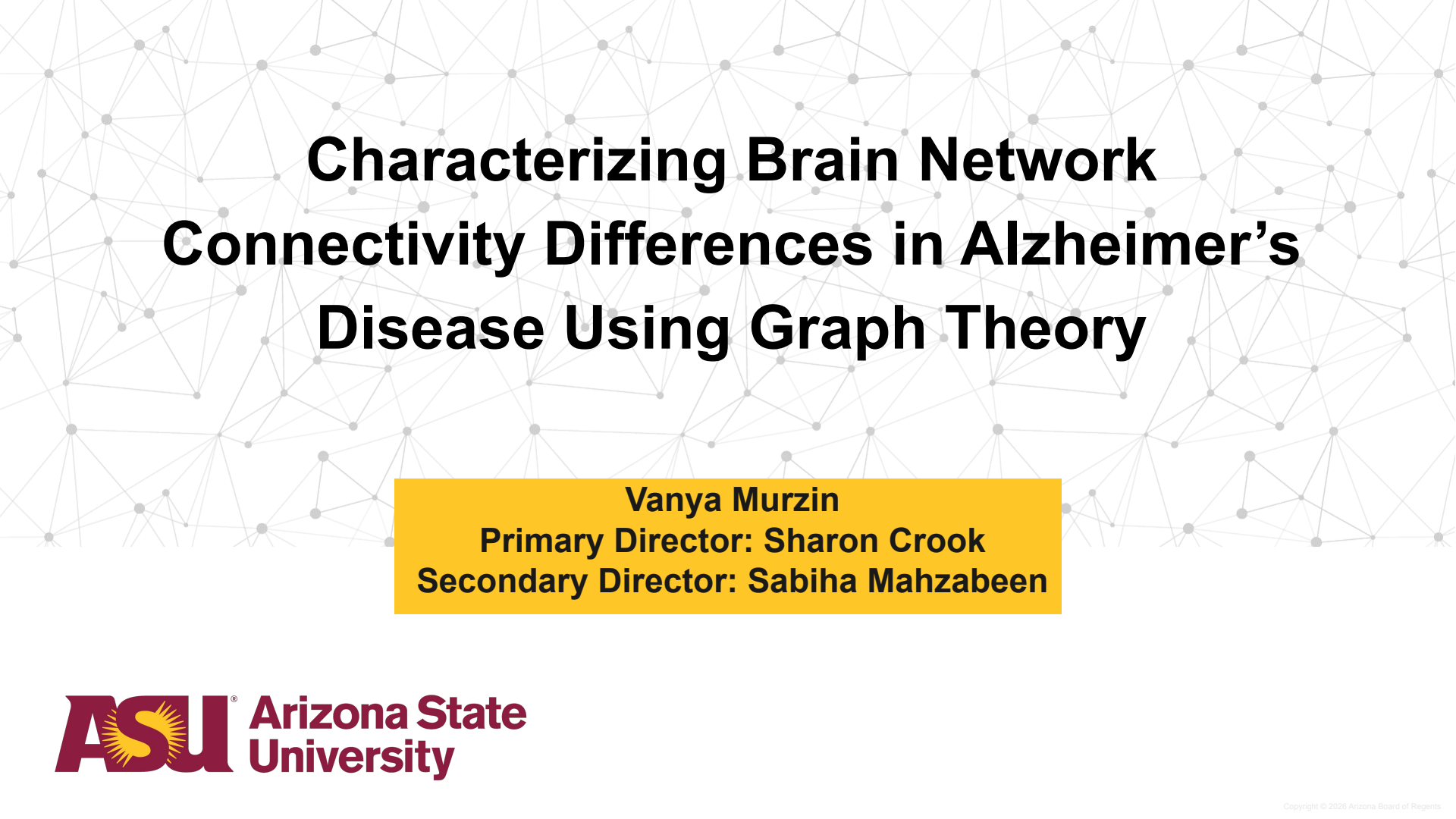


The background of the slide features a complex network graph pattern. It consists of numerous small, light gray circular nodes connected by thin, light gray lines, creating a web-like structure that fills the entire background.

Characterizing Brain Network Connectivity Differences in Alzheimer's Disease Using Graph Theory

Vanya Murzin

Primary Director: Sharon Crook

Secondary Director: Sabiha Mahzabeen

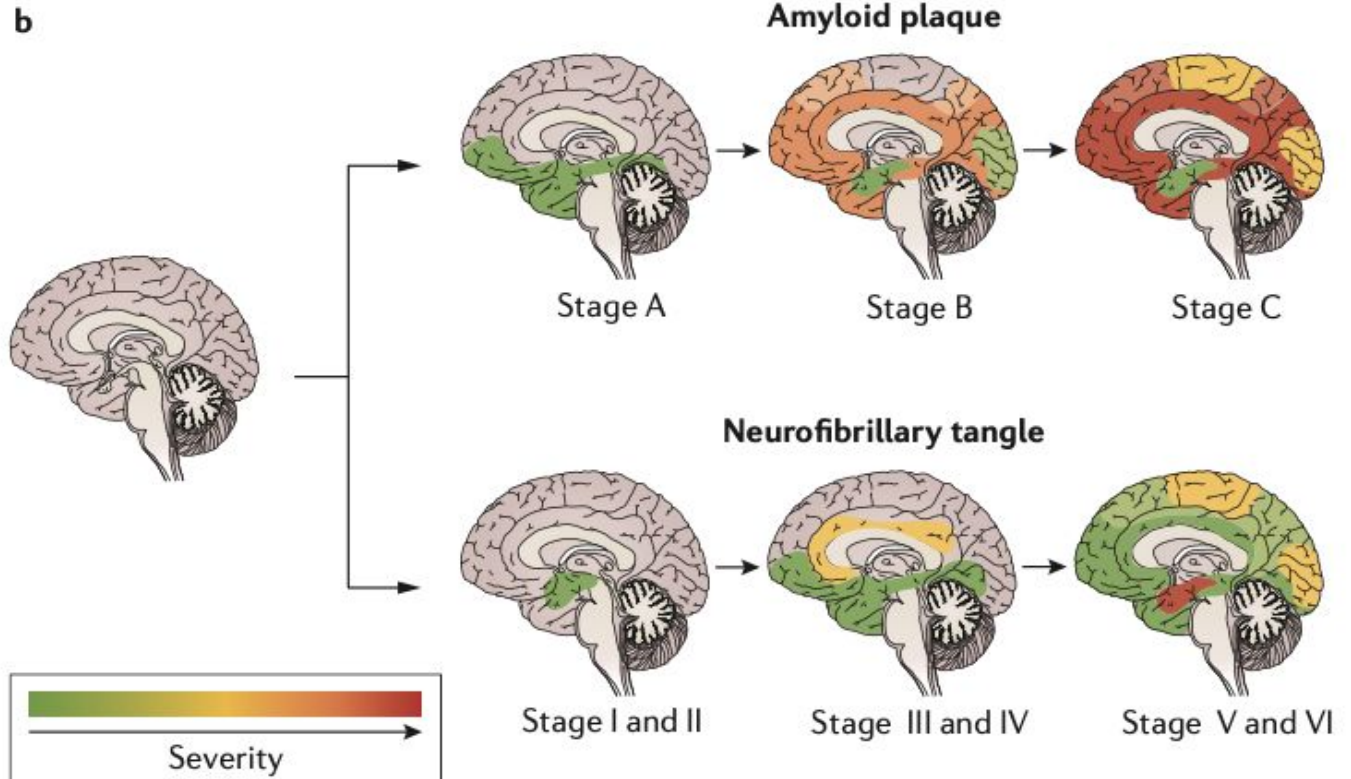
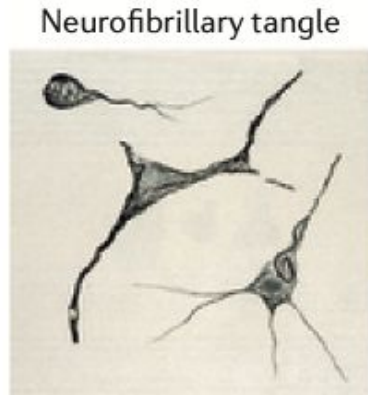
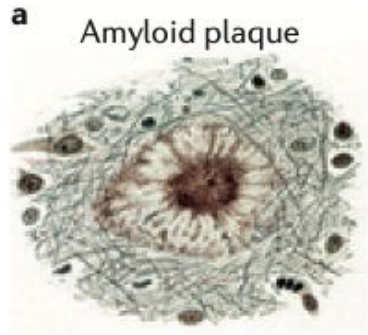
Research Questions

1. Can graph theory analysis of brain networks characterize differences in connectivity between brain regions in Alzheimer's patients versus control?
2. Can graph theory measures be used as features in a ML model to classify Alzheimer's disease?

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Alzheimer's Disease



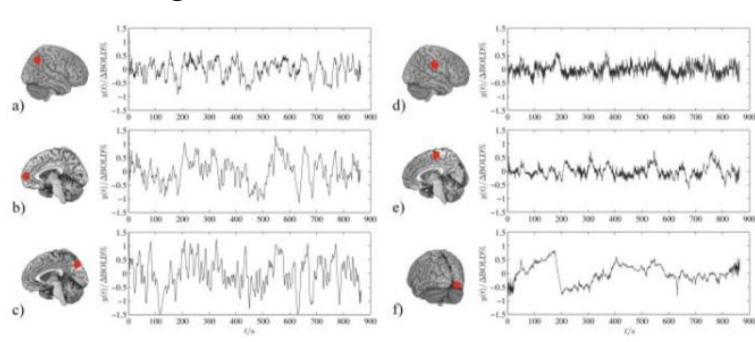
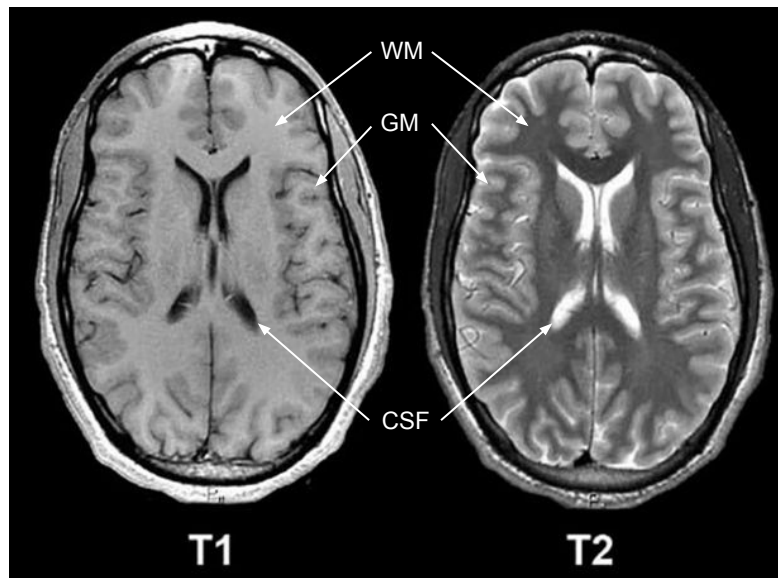
Study Design

Group	Alzheimer's Disease (AD) n = 20	Cognitively Normal (CN) n = 20
Age (years)	70–90	70–90
Age Groups	70–75, 75–80, 80–85, 85–90	70–75, 75–80, 80–85, 85–90
Sex	F (n = 10) M (n = 10)	F (n = 10) M (n = 10)

* age- and sex-matched controls

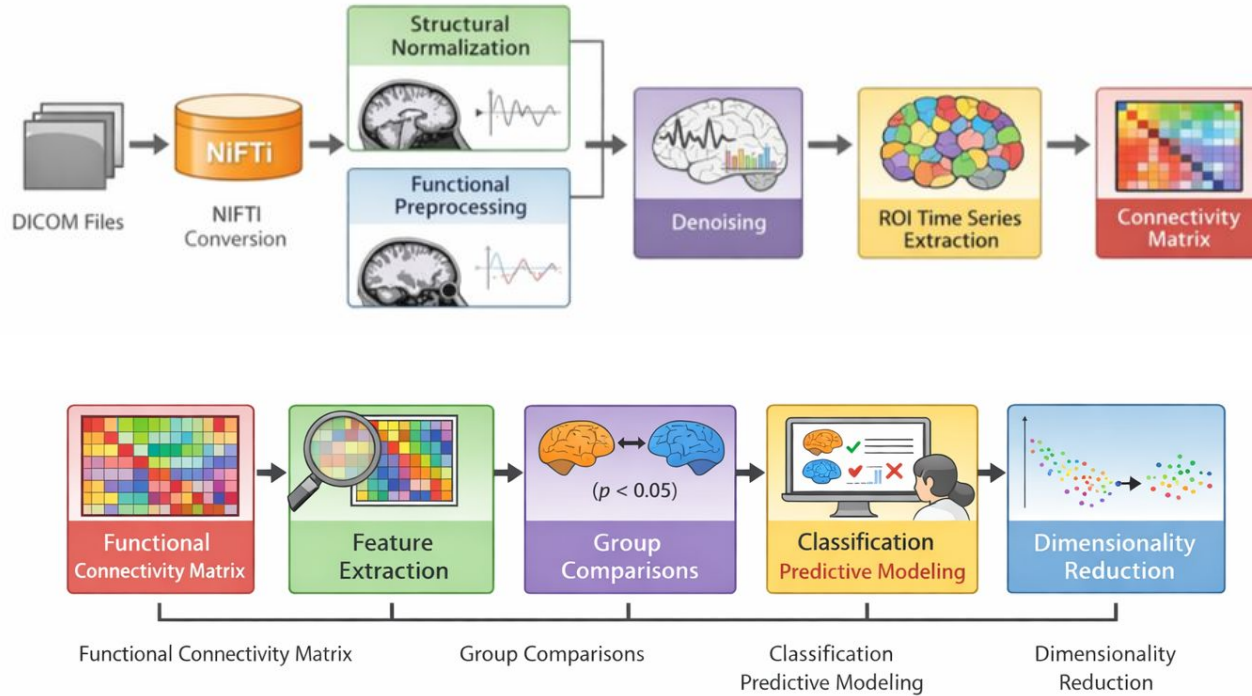
Image Data

- Structural MRI
 - T1: used to define brain anatomy and regions
 - T2: highlights pathology and fluid
- Functional (fMRI)
 - Measures changes in blood flow and oxygenation based on neuronal activity
 - BOLD (blood oxygen level dependent) signal

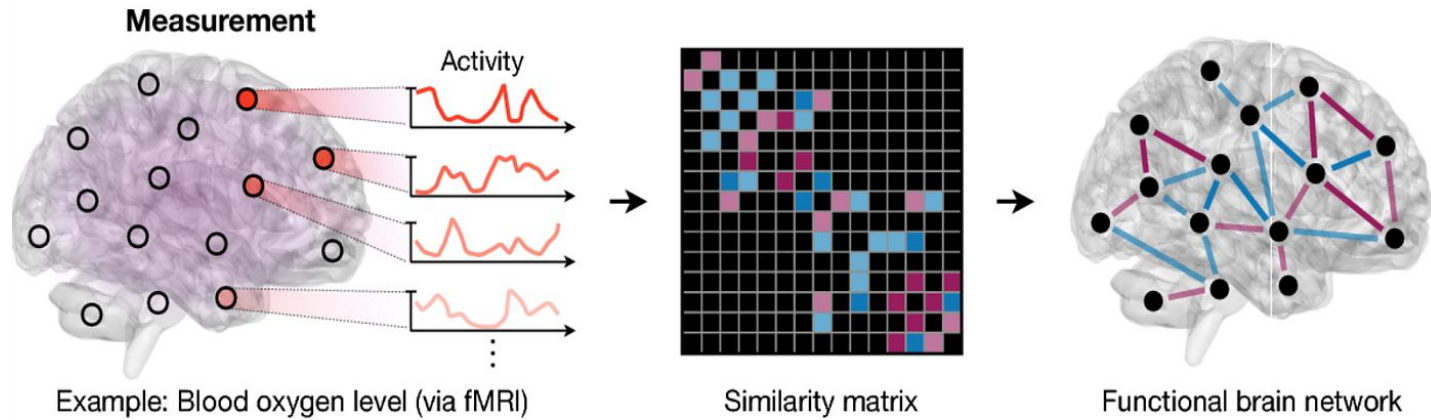


Tanenbaum et al., American Journal of Neuroradiology (2017)
Minati et al. (2015)

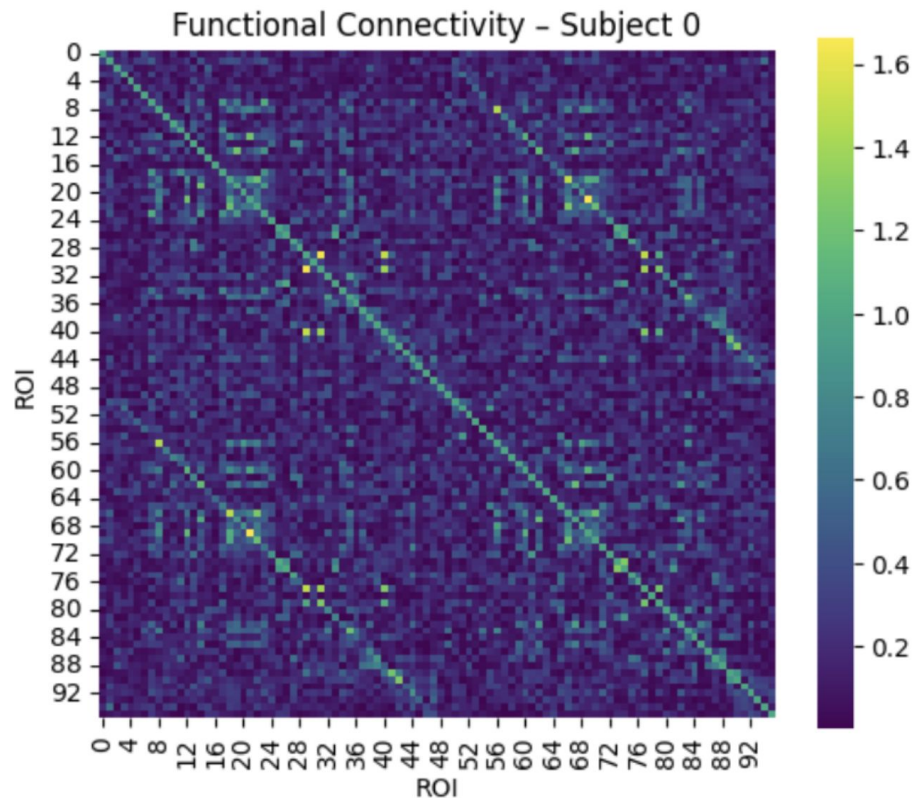
Analysis Workflow



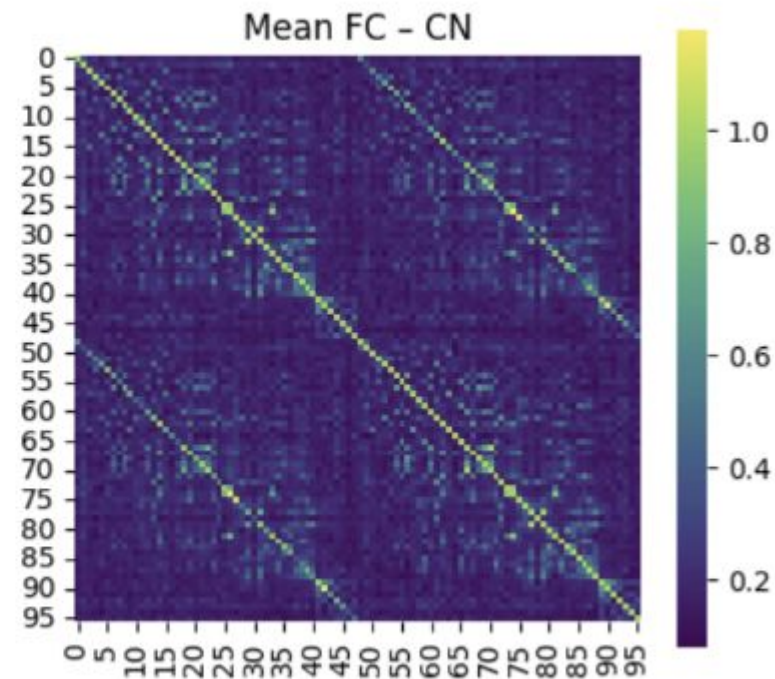
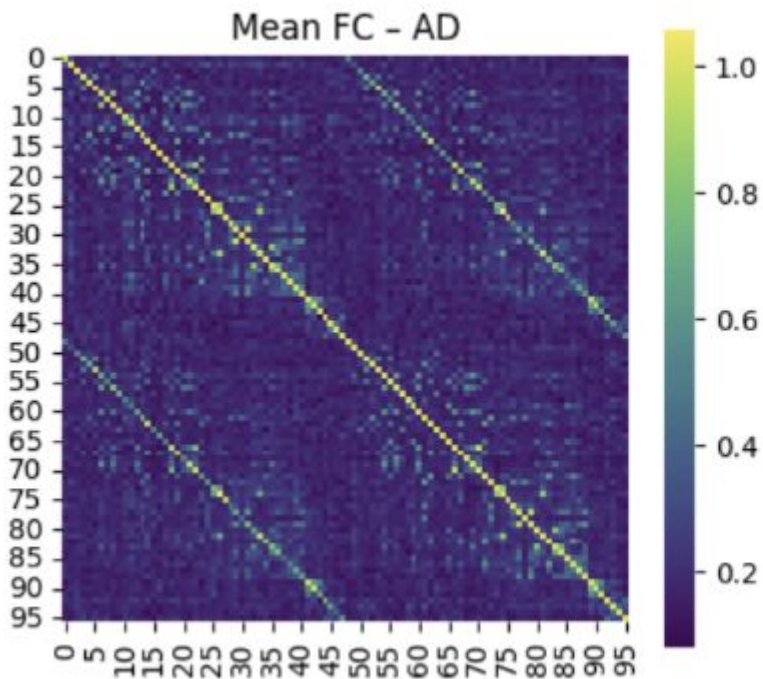
Transforming fMRI Data to Create a Functional Brain Network



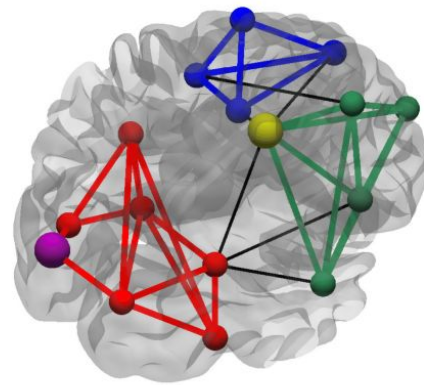
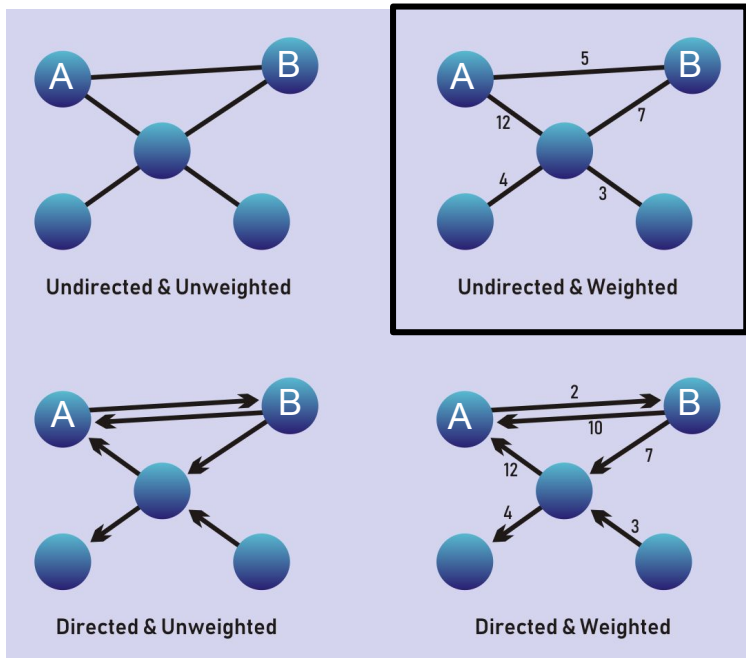
Representative Functional Connectivity Matrix



Averaged Functional Connectivity Analysis



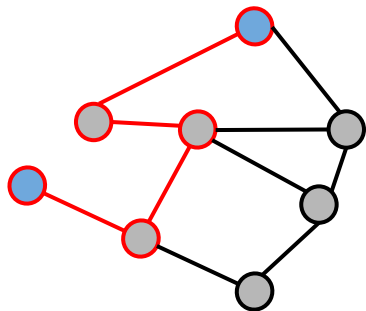
Using Graph Theory to Model Connectivity Between Brain Regions



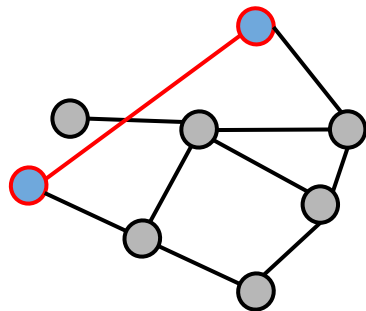
node ~ brain region
edge ~ connections between regions

Global Graph Theory Measures

Characteristic Path Length

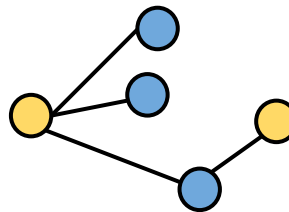


High: Many path lengths between nodes

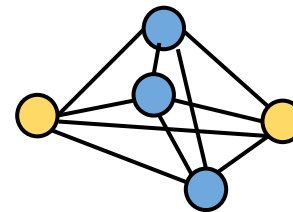


Low: One path length between nodes

Global Efficiency



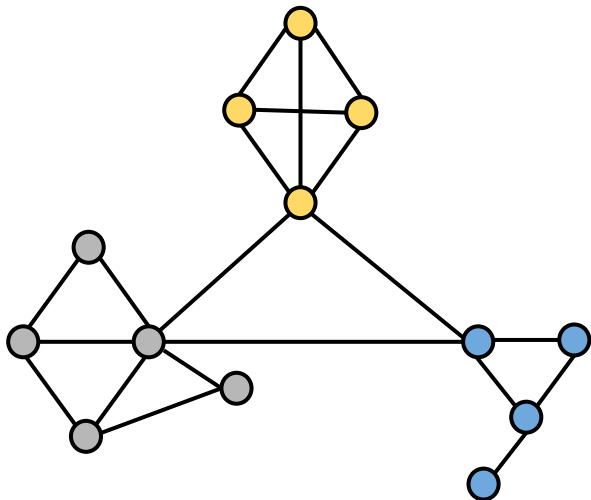
Low: Long paths, inefficient communication



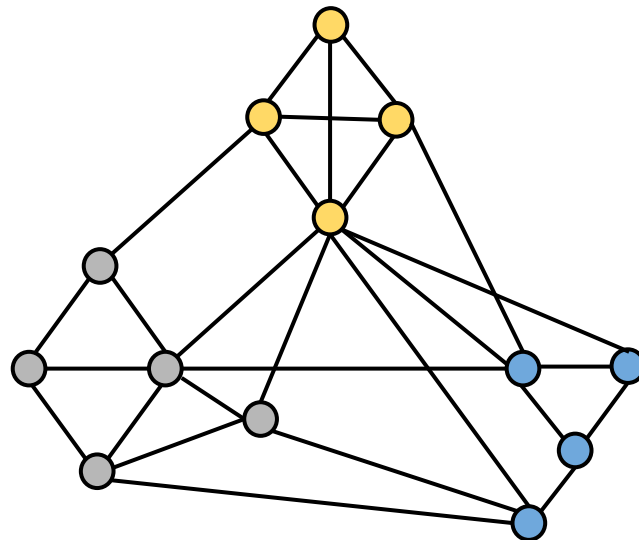
High: Short paths, efficient communication

Global Graph Theory Measures

Modularity



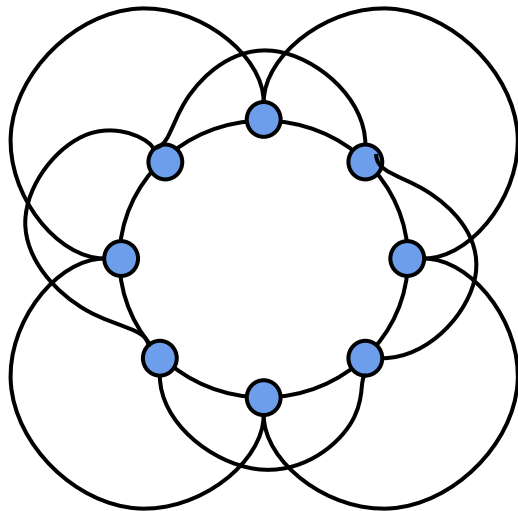
High: Dense communities efficiently connected



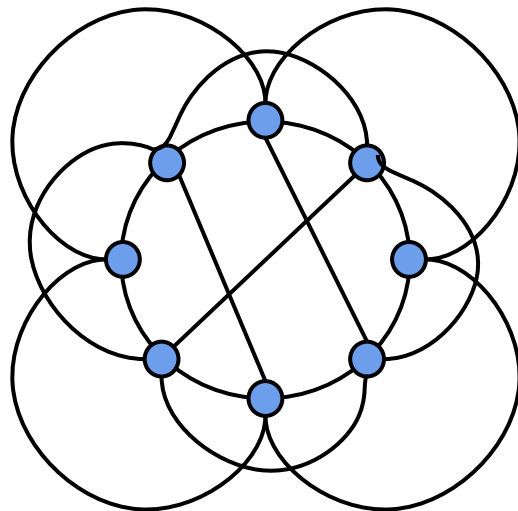
Low: Dense communities inefficiently connected

Global Graph Theory Measures

Small Worldness



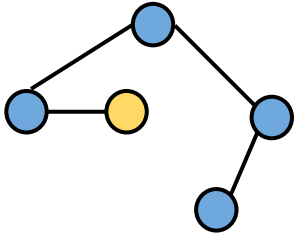
Normal Network: Locally clustered, slow long range connections



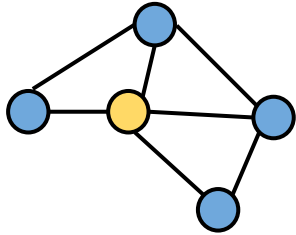
Small World Network: Locally clustered, fast long range connections

Local Graph Theory Measures

Local Efficiency

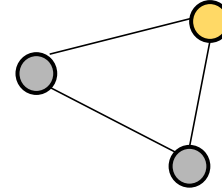


Low: One neighboring node

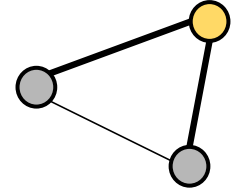


High: Many neighboring nodes

Local Strength



Low: Weak connections



High: Strong connections

Local Graph Theory Measures

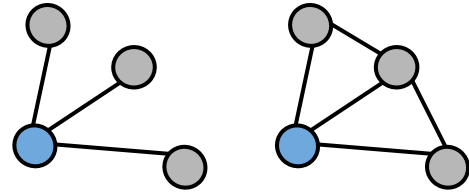
Betweenness Centrality



High: Between
other 2 nodes

Low: Not
between node

Clustering Coefficient



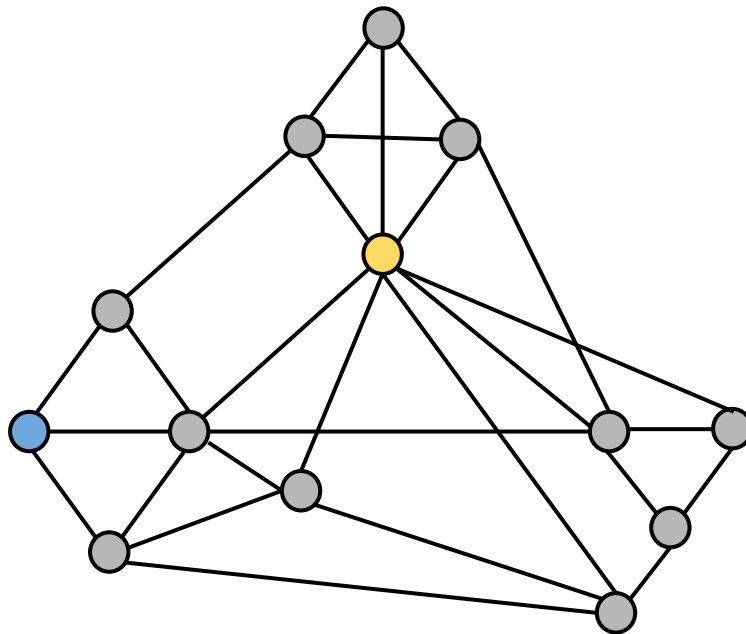
Low:
Neighbors not
connected.

High: All
neighbors
connected.

Local Graph Theory Measures

Participation Coefficient

Blue Node: Low,
participates in one
community

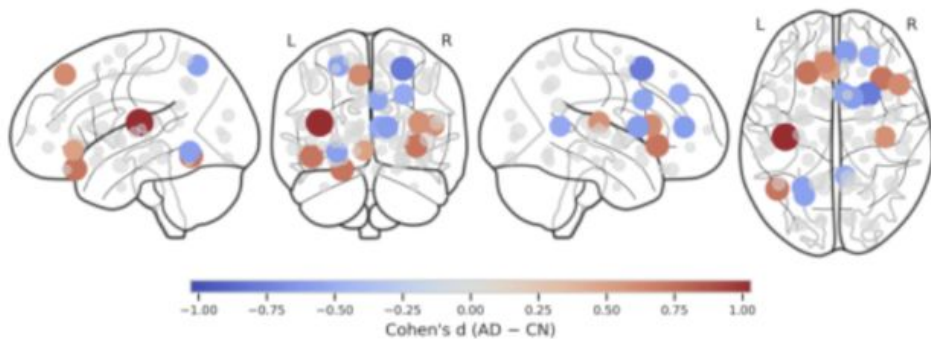


Yellow Node: High,
participates in multiple
communities

Spatial Mapping of Group Differences

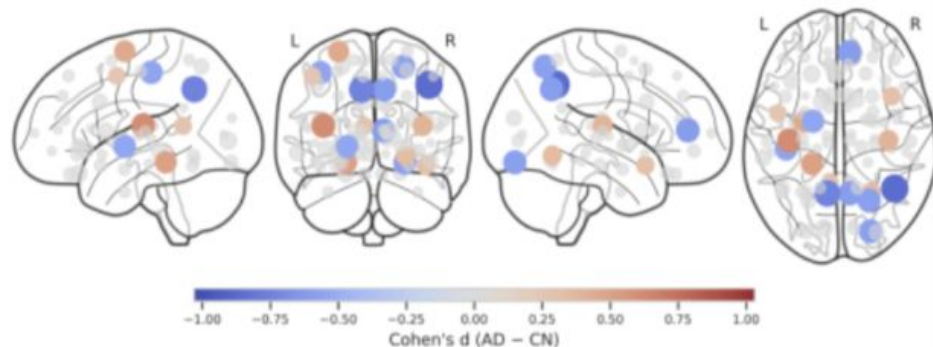
topN_both_betweenness

betweenness — Top 8 positive/negative Cohen's d



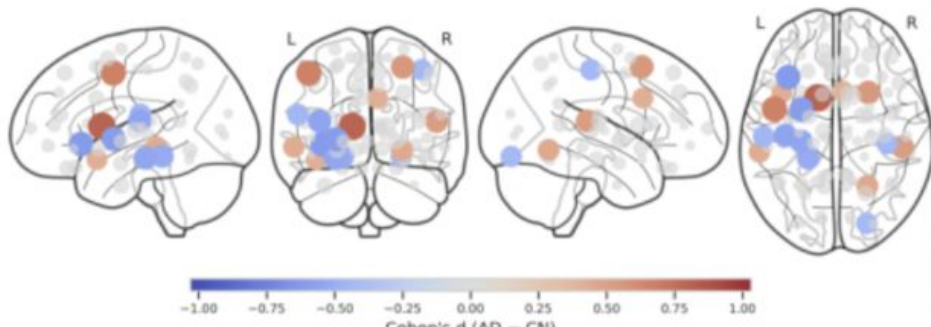
topN_both_participation

participation — Top 8 positive/negative Cohen's d



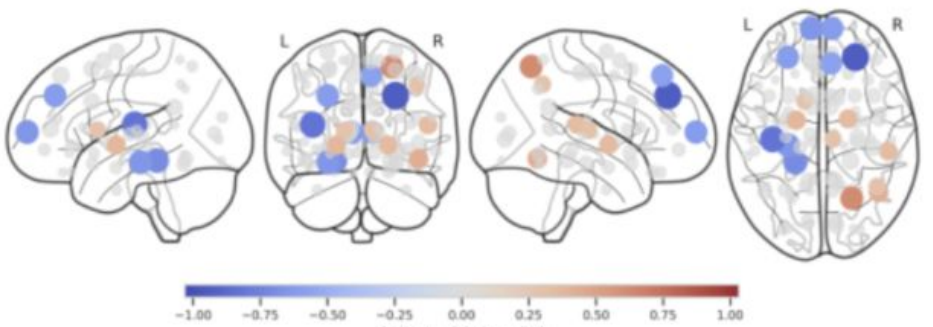
topN_both_local_efficiency

local_efficiency — Top 8 positive/negative Cohen's d



topN_both_clustering

clustering — Top 8 positive/negative Cohen's d



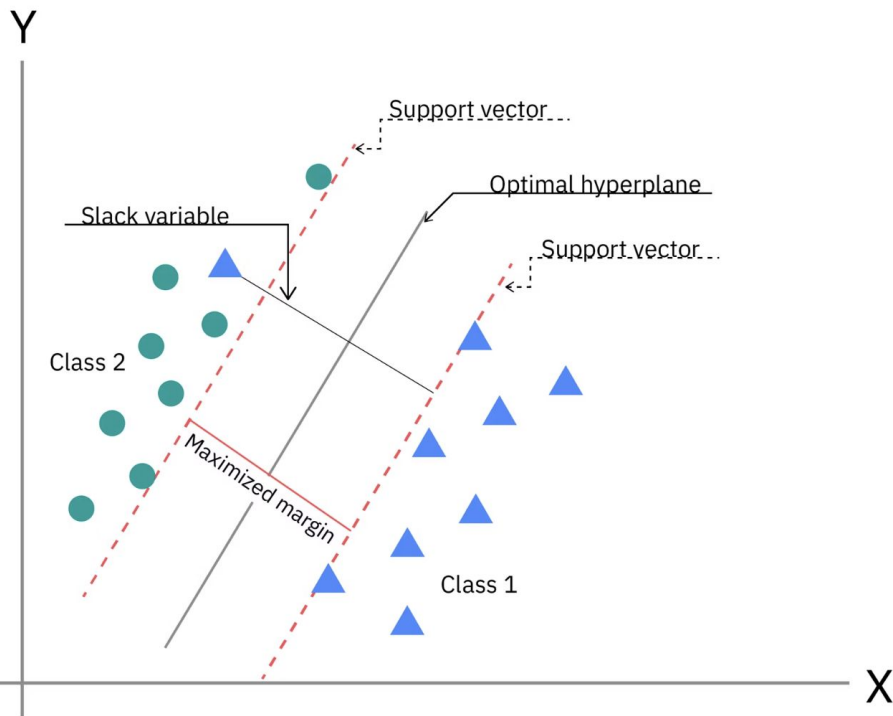
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Potential Machine Learning Models

Model	Benefits	Disadvantages	Best Use Case
 SVM	✓ High accuracy; Handles nonlinear patterns	⚠ Hard to tune; Less interpretable	🎯 Small datasets with complex structure
 Logistic Regression	✓ Simple; Fast; Interpretable	⚠ Linear only; Limited flexibility	🎯 Baseline modeling
 Random Forest	✓ Robust; Captures interactions	⚠ Less transparent; Biased to strong features	🎯 Exploratory connectivity analysis
 XGBoost	✓ High performance; Strong regularization	⚠ Many parameters; Risk of overfitting	🎯 High-accuracy clinical classification

Support Vector Machine (SVM) for Classification of Brain Network Features



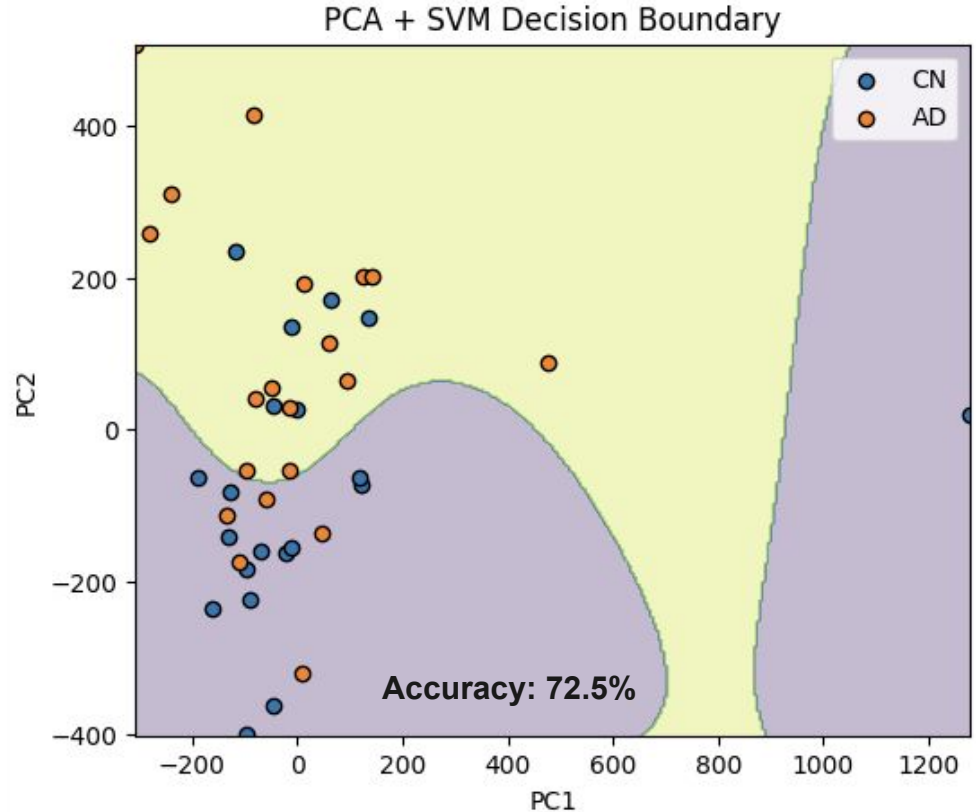
SVM is a supervised machine learning algorithm that separates groups by finding an optimal decision boundary

- Maximizes the margin between classes (AD vs CN)
- **Input:** graph theory measures
- **Output:** predicted class (AD vs CN)

SVM Successfully Predicts Class Labels

Specifications

- 40 subjects / 402 features
 - Reduced to 2D principal component space for visualization
- 5-Fold Cross Validation
 - 5 subsets of data
- GridSearchCV
 - Found that $C=10$ is optimal



Limitations

- Small sample size ($n=40$)
 - Limited statistical power; risk of overfitting
- Fixed sparsity threshold ($\text{thr}=0.3$)
 - Less biologically meaningful
- Removal of negative correlations in denoising step
 - Loss of potentially meaningful information
- Static functional connectivity
 - Ignores dynamics of brain activity
- Moderate machine learning performance
 - Insufficient for clinical application

Next Steps & Future Directions

- Can longitudinal data be used to predict AD?
 - Same patient → pre- and post-diagnosis
- Model brain regions as weighted and directed networks
- Incorporate feature selection/extraction
- Test different ML models and compare accuracy



Acknowledgements

Thank you to my primary director, Dr. Crook for helping me throughout this project and throughout all the classes I took with you.

Thank you to Kira, Dominique, Kate, and Dr. Crook for guiding and helping me throughout the whole project.

Thank you to Professor Mahzabeen for agreeing to be my second director and for teaching me everything I know about data science.



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Sharon Crook

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Kira Koyah

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Jaelyn Darden

Liyah Coleman

Eric Dudebout

Dominique Hughes

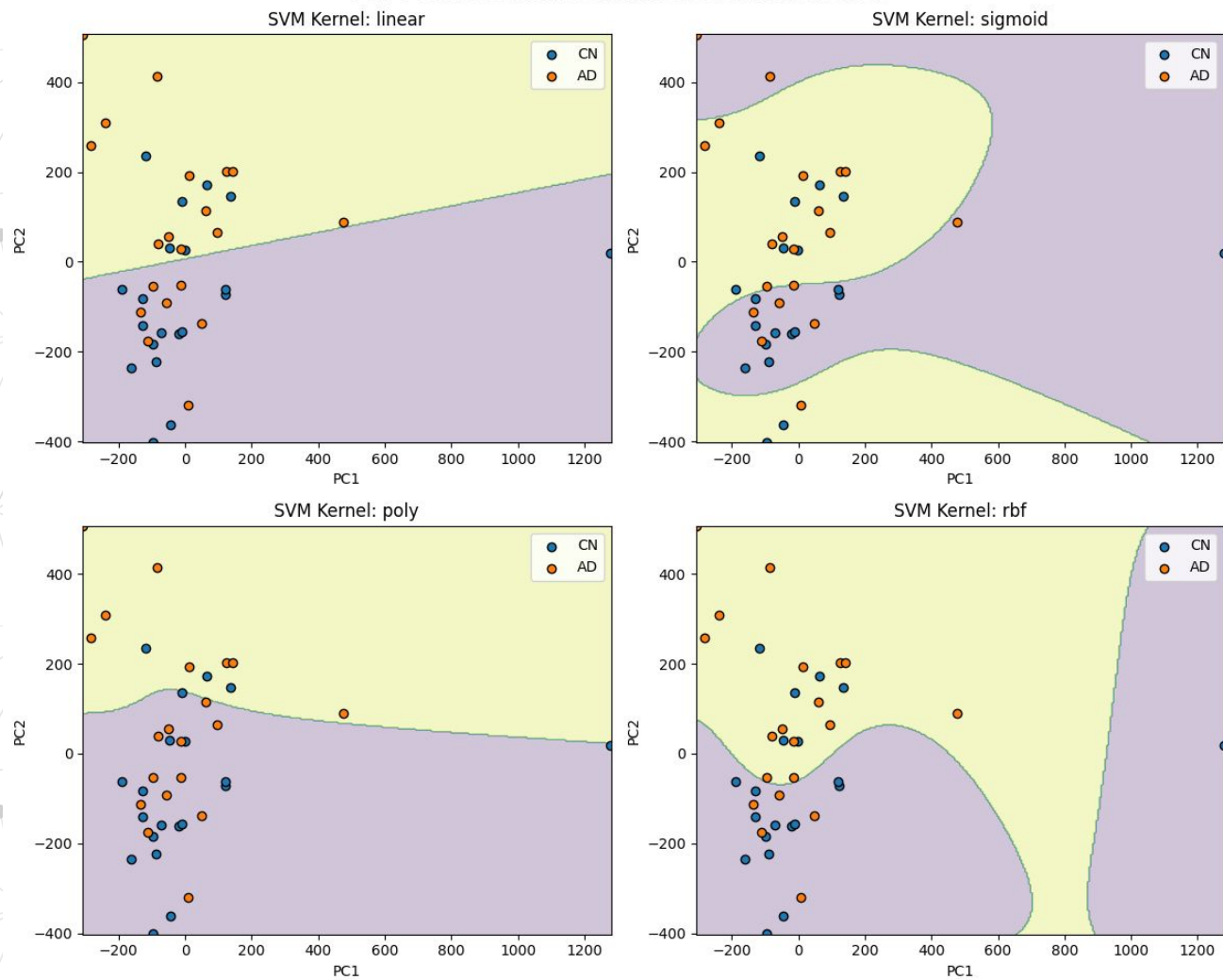
Kate Doxey



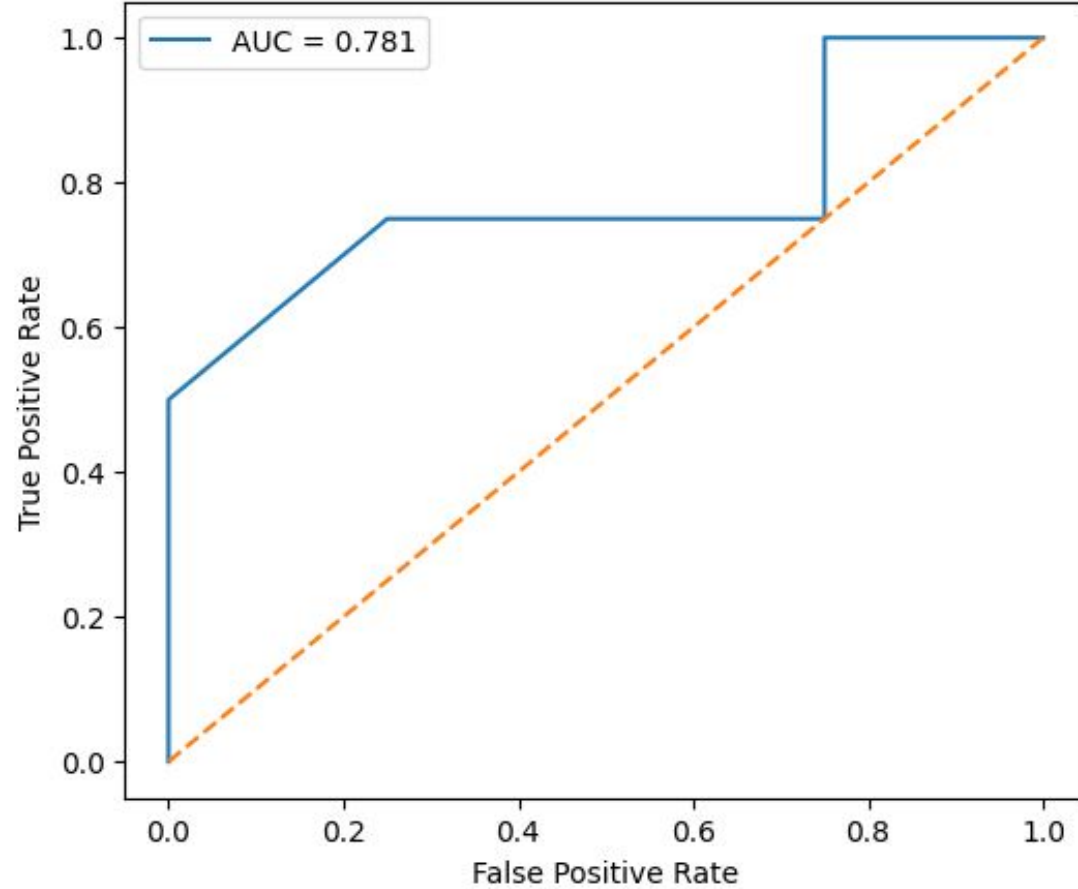
Thank you!

Questions?

PCA + SVM Decision Boundaries for Different Kernels

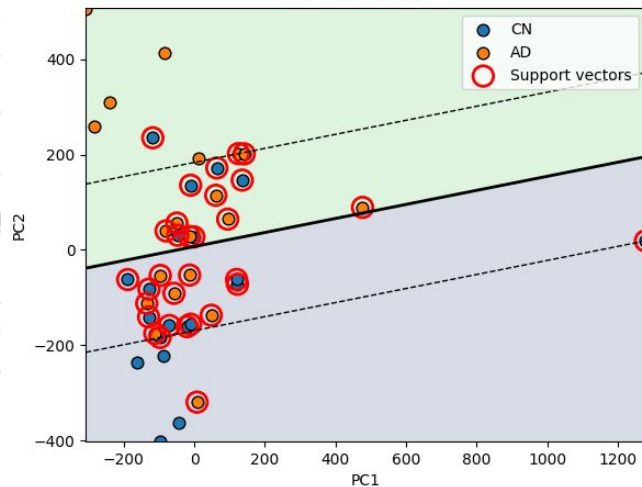


ROC Curve

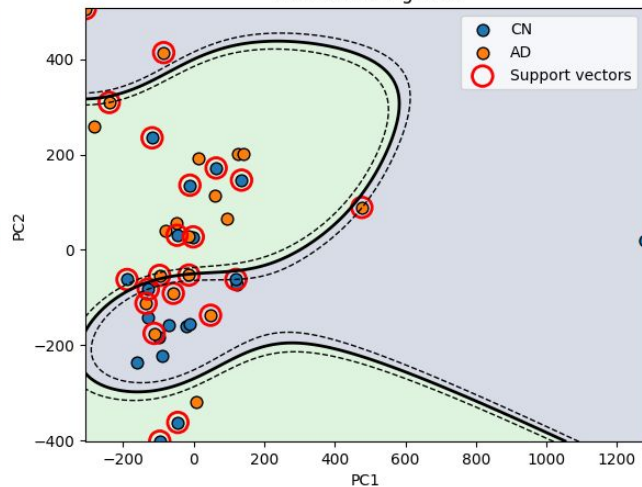


PCA + SVM Decision Boundaries with Support Vectors and Margins

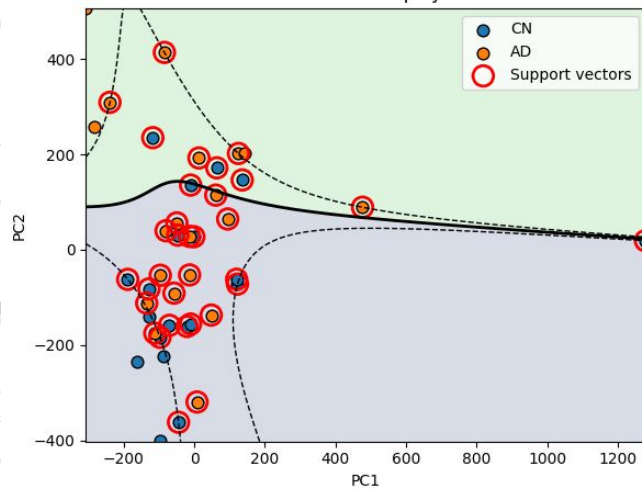
SVM Kernel: linear



SVM Kernel: sigmoid



SVM Kernel: poly



SVM Kernel: rbf

